

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

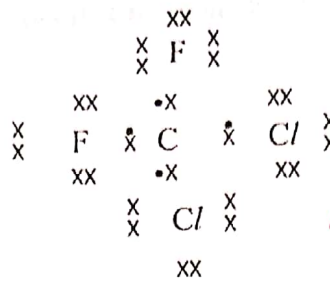
MARKING SCHEME

NOVEMBER 2019

CHEMISTRY

6031/3

1 (a) (i)



notation for Cl should be different for F and C.

- (ii) Four bonded pairs, no lone pair;
Tetrahedral;

[1]
[1]

(b) (i) Mass of carbon = $\frac{0.176}{44} \times 12 = 0.048 \text{ g}$,
 Mass of hydrogen = $\frac{0.072}{18} \times 2 = 0.008 \text{ g}$;
 Mass of oxygen = $0.120 - (0.048 + 0.008)$
 = 0.064 g;

C	H	O
$\frac{0.048}{12}$	$\frac{0.008}{1}$	$\frac{0.064}{16}$
1	2	1;

Empirical formula is CH_2O ;

$$M_r = \frac{mRT}{pV} = \frac{0.148 \times 8.31 \times 333}{1.01 \times 10^5 \times 67.7 \times 10^{-6}}$$

$$= 59.89 / 59.895.$$

$$= 60;$$

$$(\text{CH}_2\text{O})_n = 60$$

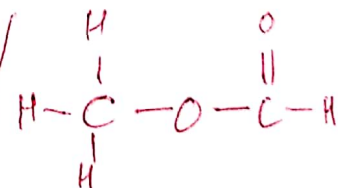
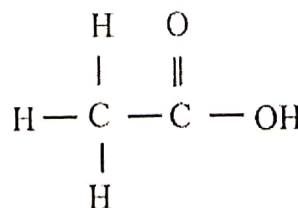
$$n = 2$$

Molecular formula is $\text{C}_2\text{H}_4\text{O}_2$;

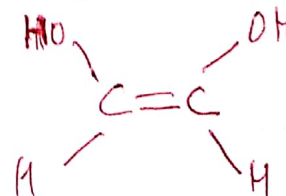
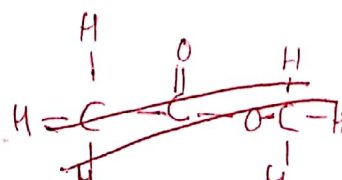
E.C.F.

[1]

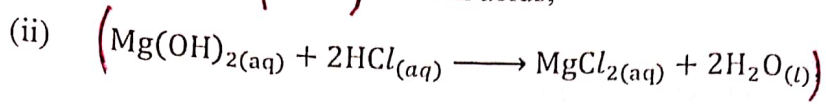
(ii)



[1]



(c) (i) Neutralises (excess) stomach acids; [1]



$$n(\text{HCl}) = \frac{0.15 \times 28.30}{4.25} \times 10^{-3} \text{ moles}; = 4.25 \times 10^{-3} \text{ moles} \quad [1]$$

E.E.E.

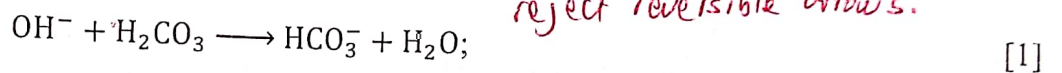
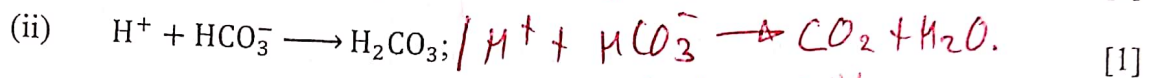
$$n(\text{Mg(OH)}_2) = \frac{1}{2} \times \frac{0.15 \times 28.30}{100} \times 10^{-3} \text{ moles} \quad [1]$$

$$m\text{Mg(OH)}_2 = 2.12 \times 10^{-3} \times 58.3 \quad [1]$$

$$= 0.124 \text{ g}$$

$$\% \text{Mg(OH)}_2 = \frac{0.124}{0.5} \times 100 = 24.8\% \quad [1]$$

2 (a) (i) regulates blood pH; / A.W maintains pH of blood. [1]



(iii) $\text{pH} = \text{pKa} + \log \left(\frac{[\text{NaHCO}_3]}{[\text{H}_2\text{CO}_3]} \right)$

E.C.F. for
pKa value.

$$5.94 = 6.33 + \log \left(\frac{x}{0.5} \right); \quad 6.337 \quad [1]$$

$$[\text{NaHCO}_3] = 0.2035 \text{ mol/dm}^3; \quad 0.2035 / 0.199 \quad [1]$$

$$n(\text{NaHCO}_3) = \text{C.V} = 0.2035 \times 250 \times 10^{-3}; \quad 2.004 / 0.199$$

$$= 0.050875 \text{ moles}; \quad 0.050875 / 0.0501 \quad [1]$$

$$m(\text{NaHCO}_3) = n.M = 0.050875 \times 84$$

$$= 4.27 \text{ g}; \quad 4.1799 / 4.2084 \text{ g} \quad [1]$$

(b) (i) $K_c = \frac{[\text{CH}_3\text{CH}(\text{OCH}_3)_2][\text{H}_2\text{O}]}{[\text{CH}_3\text{CHO}][\text{CH}_3\text{OH}]^2}; \quad [1]$

Initial	0.131	0.129	0.0785	0.0418
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Transformed	x	2x	x	x
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Equilibrium	0.131-x	0.129-2x	0.0785+x	0.0418+x
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	0.131-0.0326	0.0638	0.0785+0.0326	0.0418+0.0326
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$$= 0.0984$$

$$= 0.1111$$

$$= 0.0744$$

$$0.129 - 2x = 0.0638;$$

[1]

$$\therefore x = 0.0326$$

Correct value of x ;

All correct equilibrium moles;

$$K_c = \frac{\left(\frac{0.1111}{0.025}\right)\left(\frac{0.0744}{0.025}\right)}{\left(\frac{0.0984}{0.025}\right)\left(\frac{0.0638}{0.025}\right)^2}$$

$$= 0.516;$$

$$\frac{(\text{mol.dm}^{-3})^2}{(\text{mol.mol}^{-1})^3}$$

$$= \text{dm}^3 \text{mol}^{-1};$$

E.C.F. for
2c

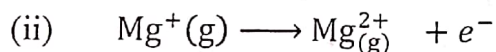
E.C.F. wrong e K values.

E.C.F. for units determined by K_c express.
Give all intermediate marks if answer is correct.

- (ii) Add concentrated H_2SO_4 a drying agent; removes water; thereby shifting the equilibrium position to the right/favouring the forward reaction;

Increase conc of CH_3CHO / Conc CH_3OH ; shifting equilibrium to the right or favouring forward rxn / decrease conc H_2O ; shifting equilibrium to the right or

- 3 (a) (i) Enthalpy change when one mole of electrons is removed from a mole of gaseous atoms to form a mole of unipositive ions; / A.W.



state symbols be there.

- (iii) There is a jump in energy between the 2nd and 3rd electron; between 10th and 11th electron; proving the 3 shells;

(b) $100 - (70 + 20.2) = 9.8\%$;

Let the relative isotopic mass of the third isotope be y ;

$$\frac{27.40 \times 70 + 28.00 \times 20.2 + 9.8 y}{100} = \text{Ar}(\text{Ar});$$

$$1918 + 565.6 + 9.8 y = 2698.2$$

$$2483.6 + 9.8 y = 2698.2$$

$$y = \frac{2698.2 - 2483.6}{9.8};$$

$$= 21.98; / 22.08$$

Award marks
for steps not
shown (c) if answer is correct.

- (i) Al_2O_3 has a high melting point; *strong ionic bonds due to high lattice enthalpy* [1]
It does not conduct electricity in (solid state); ions are not free to move; [1]
- (ii) K has a larger atomic radius *it has one extra shell than argon,* electron in outer shell less firmly held; [1]
outer most electron screened from nucleus by inner shells; [1]
- (iii) PCl_5 has more electrons per molecule; [1]
Hence greater van der Waals forces; *charges: $[\text{PCl}_4]^+ [\text{PCl}_6]^-$* [1]
can exist as ionic cpd. [Total: 15]

4

- (a) (i) 1. +5; *(A) 5+* [1]
2. +1; [1]

(ii) yellowish/green colour of Cl_2 disappears; [1]

iii $3\text{Cl}_2 + 60\text{H}^- \rightarrow 5\text{ClO}_2^- + \text{ClO}_3^- + 3\text{H}_2\text{O}$; *reject molecular eqn* [1]

(b) (i) Simultaneous oxidation and reduction; of the same species; *Aw.* [1]

(ii) $4\text{KClO}_3 \rightarrow 3\text{KClO}_4 + \text{KCl}$; [1]

(c) $\text{MCO}_3(\text{s}) \xrightarrow{\text{heat}} \text{MO}(\text{s}) + \text{CO}_2(\text{g})$; *with/without state symbols (A)* [1]

Moles of $\text{CO}_2 = \frac{0.222 \text{ g}}{44} / 5.045 \times 10^{-3} \text{ moles}$; *0.005* [1]

1 : 1 rxm ratio - moles of $\text{MCO}_3 = 5.045 \times 10^{-3}$; [1]

mass $M_r = 5.045 \times 10^{-3} \times M_r$; [1]

$\frac{0.598}{5.045 \times 10^{-3}} = \frac{0.222 \times 44}{5.045 \times 10^{-3}} (Y + 12 + 48)$; [1]

E.C.F.

$\frac{0.598}{5.045 \times 10^{-3}} = Y + 60$; [1]

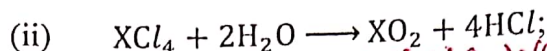
Y = 58.5 *59.6* [1]

$\text{MCO}_3 \Rightarrow \text{NiCO}_3 / \text{Ni(OH)CO}_3$; [1]

[Total: 15]

(a) (i) tetrahedral;

109.5°)

(iii) ~~X has vacant d-orbitals;~~ ~~low lying d-orbitals/has no accessible d-orbitals;~~ ~~which can allow the lone pair of an O₂ to bond datively with X;~~ ~~In CCl₄ the C atom does not have accessible d-orbitals to bond with water molecules;~~

- (b) (i) C-Cl bond energy in C-Cl₄ is ~~very short/very strong~~ ^{greater than the lead-Cl bond in PbCl₄} does not break easily / small size of carbon atom ^{hence high degree of overlap;} Carbon is ^{stable in the +4 oxidation state.}
- Pb-Cl bond energy weak/long due to large size of Pb atom; hence ^{lower degree of overlap.} PbCl₄ is less stable due to ^{inert pair effect.}
- (ii) ~~N₂ is unreactive;~~ ^{Coz of very strong N≡N bond; and weaker P-P single bonds;} short triple bonds, high bond-energy; bonds in phosphorus are weaker;
- (iii) ~~Large P atoms with vacant d-orbitals;~~ ^{Large P atoms with low lying d-orbitals; it means P. can expand its octet;} (vacant orbitals)
- (iv) ~~Halogens are non polar whereas water is polar and hence no dissolution;~~ ^{Hexane is non polar hence dissolves halogens;}

(v) Cl₂ is a stronger oxidising agent; hence oxidises I⁻ to I₂;
 $\text{Cl}_2 + 2\text{I}^- \rightarrow \text{I}_2 + 2\text{Cl}^-$

[Total:15]

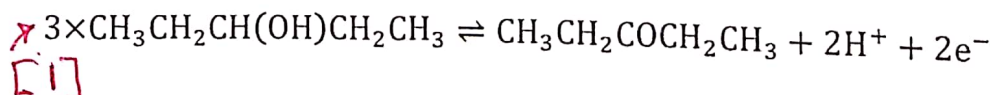
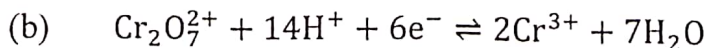
6 (a) (i) Stronger van der Waals forces in pentan-3-ol due to more electrons per molecule/larger size/AW; ^{more contact points.}

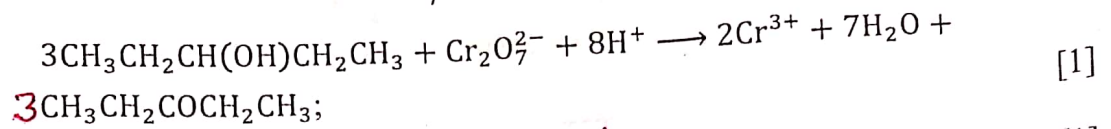
(A) stability of the conjugate base / acid pair. (ii) Propanol is ~~more acidic/AW;~~ ^{nothing to do with less acidic.}

There are two alkyl groups in pentan-3-ol which strength the O-H bond compared to only one in propan-1-ol/AW; ^{comparison.}

(iii) - add K₂Cr₂O₇ and H₂SO_{4(aq)} then heat; ^{acidified} / warm / reflux. to the product add 2, 4 DNPH; / PCl₅ only pentan-3-ol gives a yellow/orange ppt; / white fume (PCl₅)

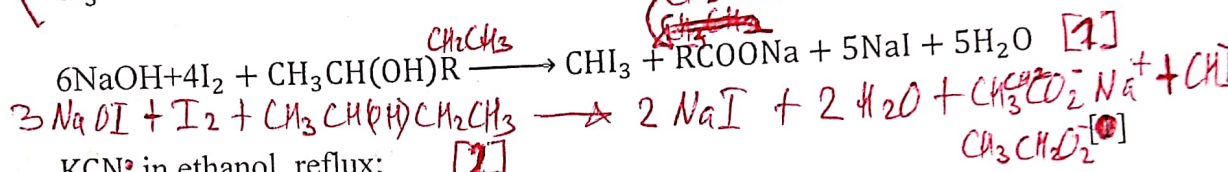
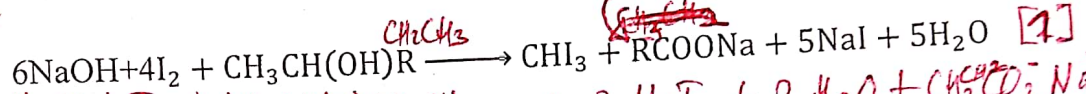
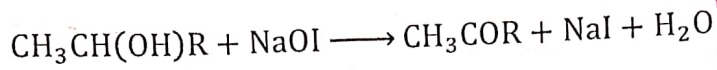
Esterification
 only propanol gives sweet smell





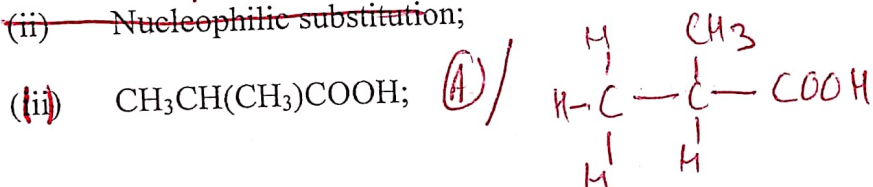
(c) (i) Yellow crystals (of CHI_3); */ppt / solid.* [1]

(ii) Oxidation; */ Redox.* [1]



(d) (i) KCN in ethanol, reflux; *NaCN / alcohol* [1]

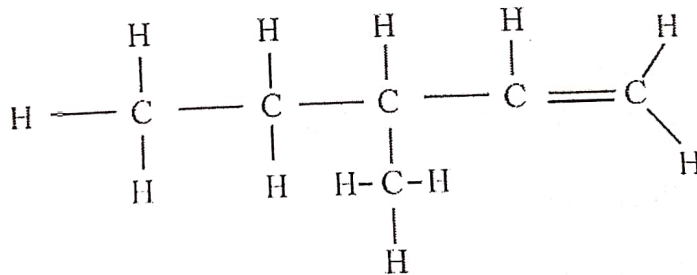
(ii) ~~Nucleophilic substitution;~~



[Total: 15]

7

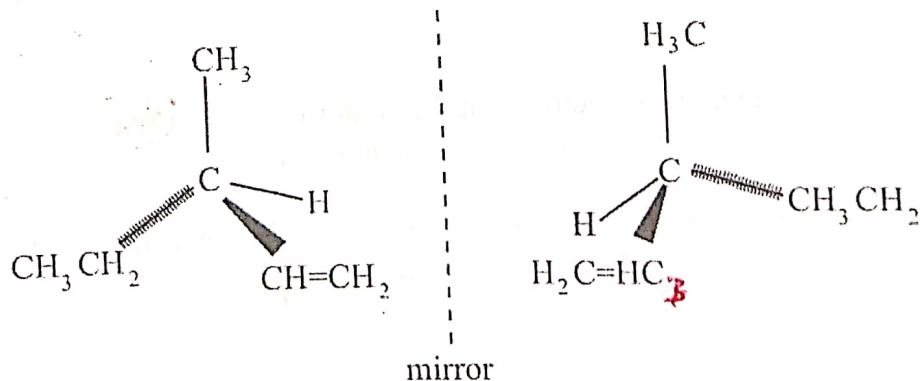
(a) (i)



[1]

(ii) 3-methylpent-1-ene; */ 3 methyl pentene* [1]

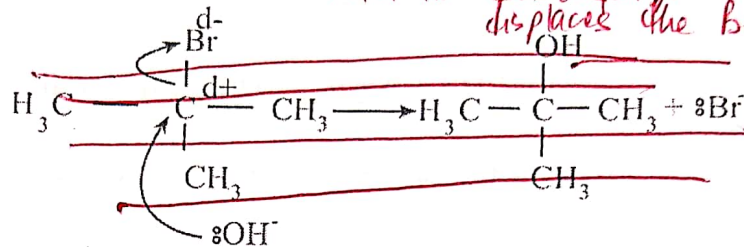
(iii)



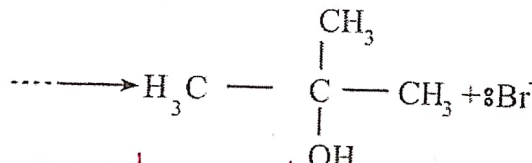
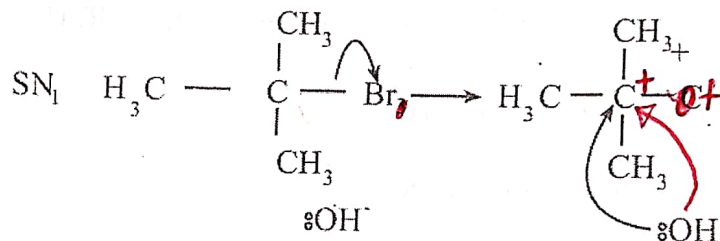
[2]

insist on 3D configuration.

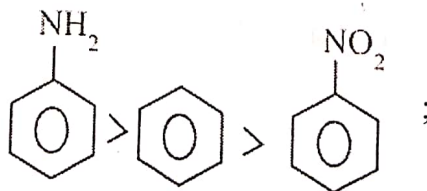
- (b) (i) 1. ~~KOH/NaOH~~ (aq), reflux; [1]
 2. ~~Ethanoic~~ (NaOH/KOH) reflux; *ethanolic and reflux;* [1]
 (ii) Reaction 1 Nucleophilic substitution; *(A) SN₁ or SN₂* [1]
Reaction 2 elimination
 (iii) ~~Reaction II~~ elimination; *Rxn I - nucleophile; Base/Catalyst; provide OH that displaces the Br* [1]



b(ii)



- 8 (a) (i) *reagent condition.*
 concHNO₃, concH₂SO₄; and 50 °C ≤ T ≤ 60 °C; *Temperature.* [1]
 (ii) Reduction; [1]
 (iii) Reactivity decreases in the order; [1]



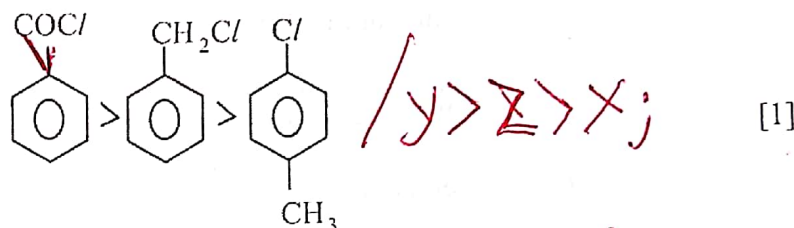
lone pair of electrons on the N atom in NH₂ partially overlaps/delocalises into the ring increasing the charge density;

the electronegative oxygen atoms in nitrobenzene withdraw electrons from ring reducing the charge density;

- (b) $\text{CH}_2\text{Cl}_2\text{COOH} > \text{CH}_2\text{ClCOOH} > \text{CH}_2\text{COOH}$; [1]
 The higher the K_a value, the stronger the acid; [1]
 due to the electron withdrawing effect/ negative inductive effect of Cl;

which stabilises the anion/ RCO_2^- OR which spreads the negative charge more OR which weakens the O-H bond OR which facilitates the H^+ donation OR which makes the equilibrium $\text{RCO}_2\text{H} \rightleftharpoons \text{RCO}_2^-$ lie further to the right [1]
(CH₃ group strengthens OH bond)

- (c) (i) electrophilic substitution; [1]
 (ii) UV light; [1]
 (d)



Ease of hydrolysis decreases in the shown order

~~The higher the K_a value, the stronger the acid;~~ [1]

$Y \rightarrow$ Has a highly δ^+ carbon atom bonded to two highly electronegative atoms; */AN* [1]

$Z \rightarrow$ Has one electron withdrawing group) [1]

for X $\rightarrow$ Lone pair of electrons on the Cl atom delocalises over the benzene ring which makes the C-Cl bond difficult to break/ C-Cl bond has a partial double bond character; [1]

OR delocalised electrons on the ring repel the OH^-

[Total:12]

9

- (a) Mn^{2+} is more stable;

[1]

has a half filled 3d orbital;

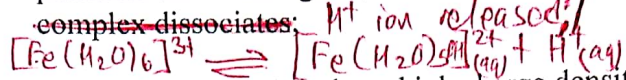
[1]

- (b) (i) high charge density on Fe^{3+} ;
polarises the water ligand;

[1]

~~complex dissociates:~~ H^+ ion released;

[1]



[1]

- (ii) highly discharged ions have high charge density;
polarises water ligands;

[1]

compounds dissociates forming oxyanions; $MnO_4^- / Cr_2O_7^{2-}$

[1]

- (c) (i) ~~A solute~~ which is soluble in two immiscible solvents distributes itself between them in such a way that the ratio of the concentration in the two solvents is a constant at a given temperature; / AW

[1]

$$K_d = \frac{[X]_{\text{solvent A}}}{[X]_{\text{solvent B}}}$$

Define the parameters

for a solute X w/c miscible in both

- (ii) Temperature; / Solvent A & B at constant temperature.
 / Conc / Chemical nature of solute in both solvents; [1]

- (iii) 1. Let x be mass of I_2

$$\frac{x}{100} \times \frac{100}{8-x} = 4$$

$$x = 6.40 \text{ g};$$

[1]

2. Successive extractions

1st extraction

$$\frac{x_1}{50} \times \frac{100}{8-x} = 4$$

$$x_1 = 5.33 \text{ g};$$

[1]

Second extraction : mass I_2 remaining = $(8 - 5.33)$ g

$$= 2.67 \text{ g}$$

$$\frac{x_2}{50} \times \frac{100}{2.67-x_2}$$

$$x_2 = 1.78 \text{ g};$$

[1]

$$= 7.11 \text{ g};$$

[1]

- (iv) Comment:

More efficient to use small portions of solvent; $7.11 \text{ g} > 6.49 \text{ g}$

[1]